



Update on the SAP GUI Family

Q1/2023

PUBLIC

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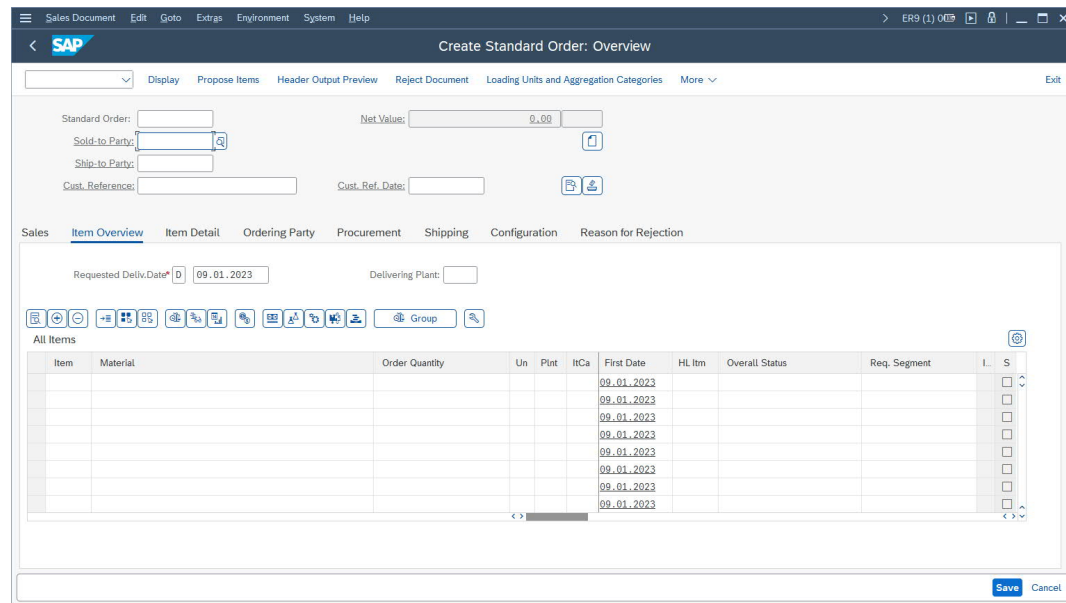
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Agenda

- What is the SAP GUI Family
- SAP GUI for Windows
- SAP GUI for HTML
- SAP GUI for Java
- SAP Business Client
- Client Integration Options

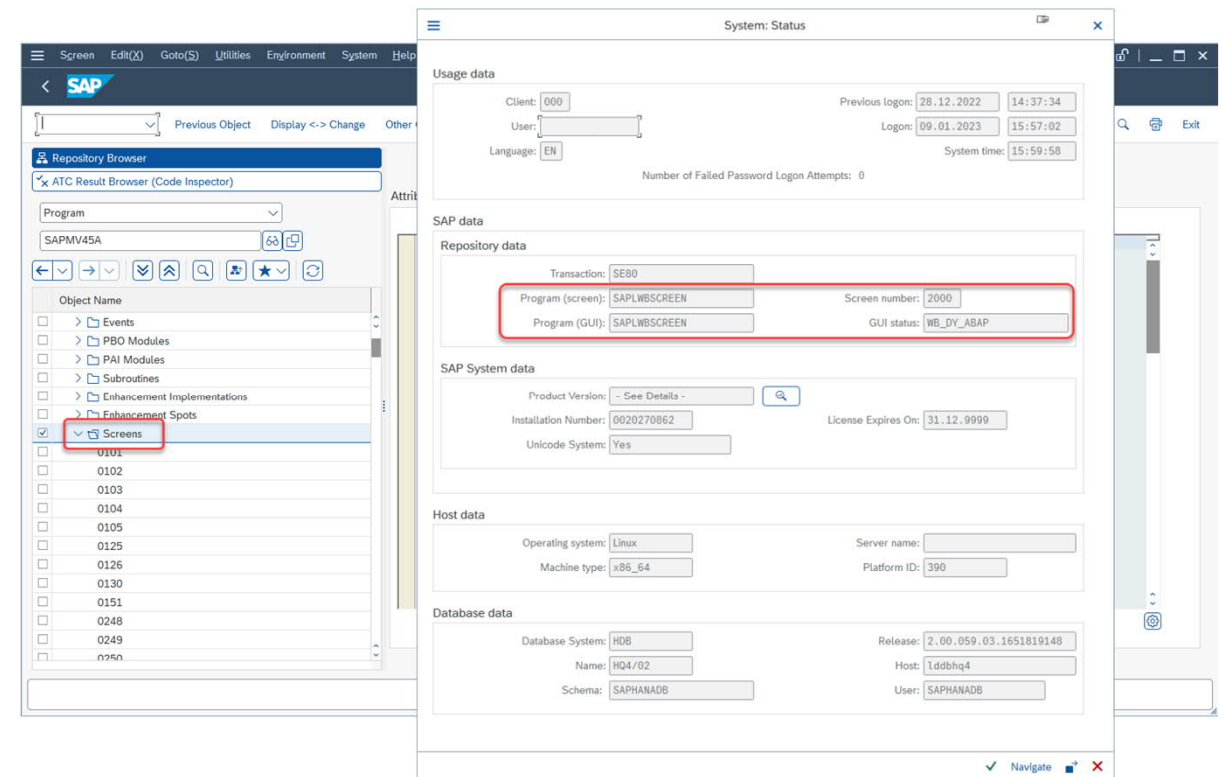
Technologies Covered in This Presentation

This presentation is about the Dynpro-based front end technologies.



In English DYNPROS are usually called screens

Dynpro is the abbreviation for DYNAmic PROgram



Screen

Edit(X)

Goto(S)

Utilities

Environment

System

Help

Previous Object

Display <-> Change

Other

Repository Browser

ATC Result Browser (Code Inspector)

Program

SAPMV45A

←

→

↕

↗

🔍

🔗

★

↻

Object Name

<

>

Events

<

>

PBO Modules

<

>

PAI Modules

<

>

Subroutines

<

>

Enhancement Implementations

<

>

Enhancement Spots

<

>

Screens

0101

0102

0103

0104

0105

0125

0126

0130

0151

0248

0249

0250

System: Status

Usage data

Client: 000

Previous logon: 28.12.2022 14:37:34

User: ANZEIGER

Logon: 09.01.2023 15:57:02

Language: EN

System time: 15:59:58

Number of Failed Password Logon Attempts: 0

SAP data

Repository data

Transaction: SE80

Program (screen): SAPLBSCREEN

Screen number: 2000

Program (GUI): SAPLBSCREEN

GUI status: WB_DY_ABAP

SAP System data

Product Version: - See Details -

Installation Number: 0020270862

License Expires On: 31.12.9999

Unicode System: Yes

Host data

Operating system: Linux

Machine type: x86_64

Server name: ldcihq4_HQ4_00

Platform ID: 390

Database data

Database System: HDB

Name: HQ4/02

Schema: SAPHANADB

Release: 2.00.059.03.1651819148

Host: lddbhq4

User: SAPHANADB

✓

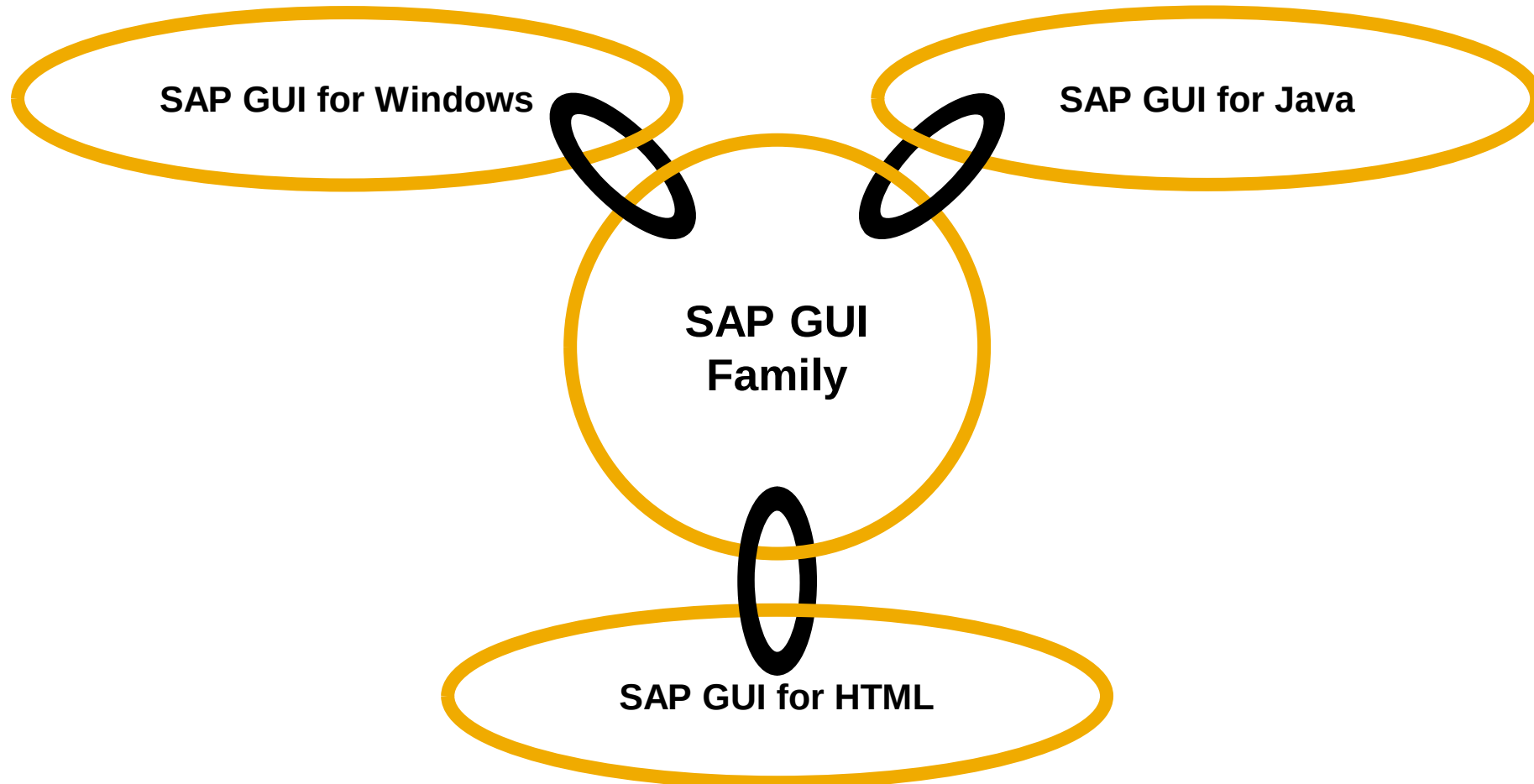
Navigate

✗

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The members of the SAP GUI Family



The Fundamentals of the SAP GUI Family

Separation of user interface and SAP system

- Changes in the GUI do not affect business rules and logic
- Business data reside in the back end system

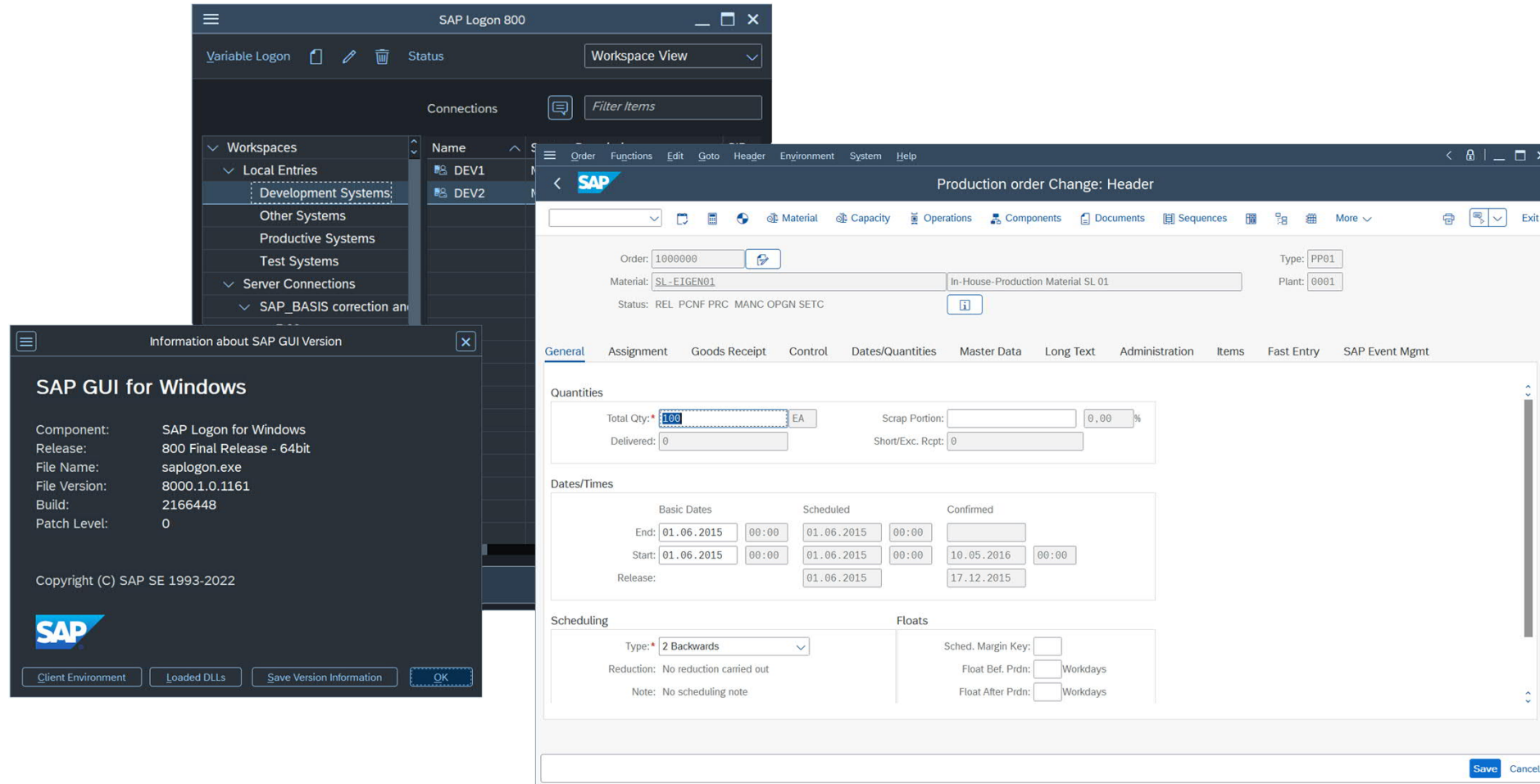
Why are there multiple GUIs for the same applications?

- Different client operating systems (Windows, Linux, ...)
- Different user experience (power user, plant worker, ...)
- Different tasks (developer, clerk, worker, ...)
- Different environments (standalone client, portal integration, ...)

Best fit into the existing infrastructure

- Low or high bandwidth available
- Use of terminal servers possible
- Available for all supported back end releases

SAP GUI for Windows



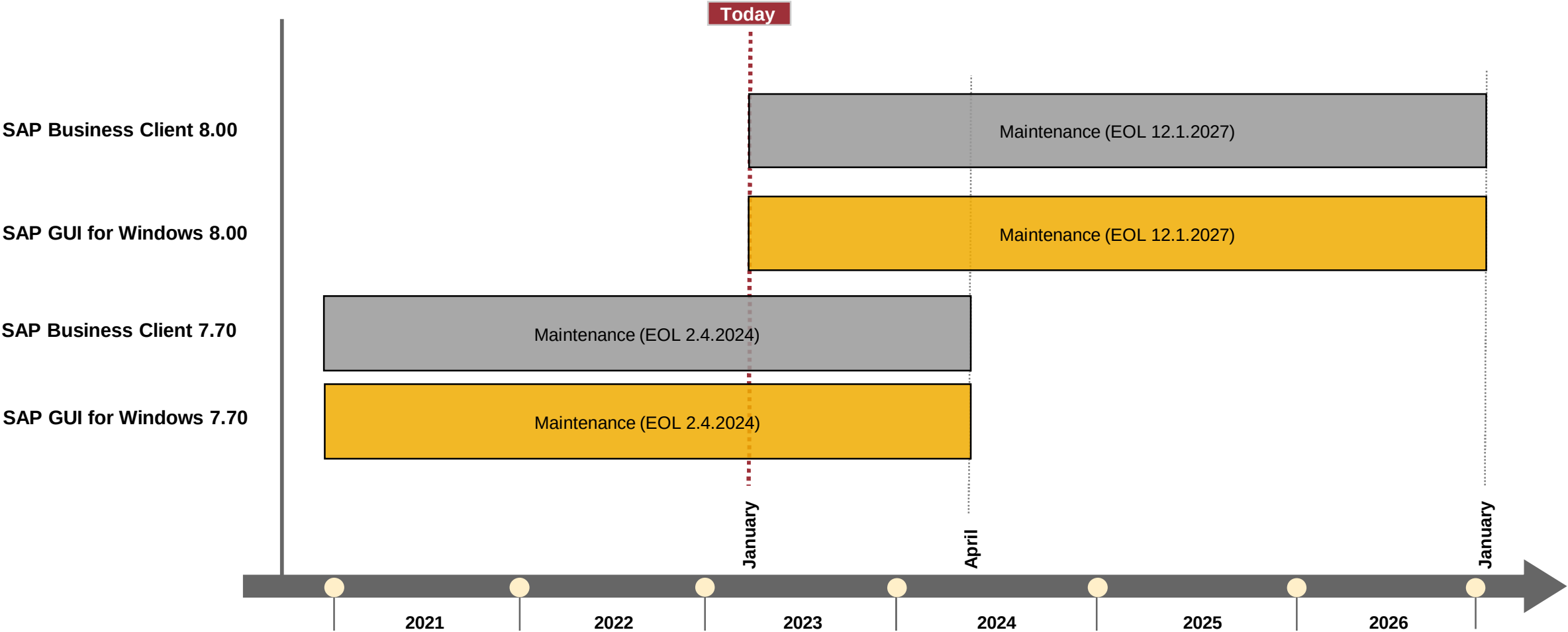
What's new in SAP GUI for Windows 8.00

Among others the following new features are included:

- **True 64bit version:** Up to release 7.70, SAP GUI for Windows was only available as a 32bit program with the exception being the automation controls (Logon Control, Table Factory Control, BAPI Control and Function Control). The 32bit version works well on 64bit operating systems running in the so-called “emulation”. However, this has a couple of drawbacks when it comes to memory utilization and integration with other 64bit processes. Therefore, as of SAP GUI for Windows 8.00, the SAP GUI delivery includes both a 32bit version and a 64bit version. You can decide which version to use (parallel usage of both versions on the same client PC is not possible). More information can be found in SAP Note [3218166](#).
- **Modernized rendering in Quartz Themes:** The rendering engine used for displaying the Quartz Themes has been completely reworked leading to a more harmonized display of Quartz (for example the rounded corners look much better). Several visual aspects like the dotted focus visualization or the display of the command field have been improved.
- **Fully usable external browser window for HTML Control based on Edge (WebView2):** The HTML Control based on WebView2 now offers the feature to open a fully usable browser window via a dedicated browser extension. In release 7.70 the HTML Control could only open a "browser tab" without navigation or menu.
- **Usability improvements:** Keyboard navigation for table controls, dialogbox containers and radiobutton groups has been extended. The combobox control now also supports showing keys and sorting by keys and the tree control offers a scrolling indicator like Table Control and ALV already did.
- **Multiple minor enhancements:** Branding images can be configured per system and client, sorting and ordering of columns in SAP Logon has been improved, SAP Logon creates backups of the user's SAP UI landscape files and much more.

For more information refer to SAP Note [3075781](#).

Support timeline



Refer to [SAP Note 2302074 \(SAP Business Client\)](#) and [SAP Note 147519 \(SAP GUI for Windows\)](#) for details

Supported operating systems and Office versions

SAP GUI for Windows 8.00 is supported on

- Windows 10 (Business Editions), 32 and 64 (AMD64, EM64T) bit versions
- Windows 11 (Business Editions), 32 and 64 (AMD64, EM64T) bit versions
- Windows Server 2016 64 (AMD, EM64T) bit versions
- Windows Server 2019 64 (AMD, EM64T) bit versions
- Windows Server 2022 64 (AMD, EM64T) bit versions

Details can be found in [SAP Note 66971](#).

Microsoft Office Support (in SAP Desktop Office Integration)

- Office 2016 (32 and 64 bit)
- Office 2019 (32 and 64 bit)
- Office 2021 (32 and 64 bit)
- Since Microsoft Office 365 ProPlus (msi or Click2Run) installs Office 2016 or newer it is also supported

Details can be found in SAP Note [722513](#).

SAP GUI for HTML

Create Standard Order: Overview

Standard Order: Net Value:

Sold-To Party: Ship-To Party:

Cust. Reference: Cust. Ref. Date:

Sales Item Overview Item Detail Ordering Party Procurement Shipping Configuration Reason for Rejection

General header data

Description:

* Req. Deliv. Date: Deliver Plant:

Complete Div.: ☐ Total Weight:

Delivery Block: Volume:

Billing Block: Pricing Date:

Total Amount: Doc. Currency:

Pyl Terms:

All Items

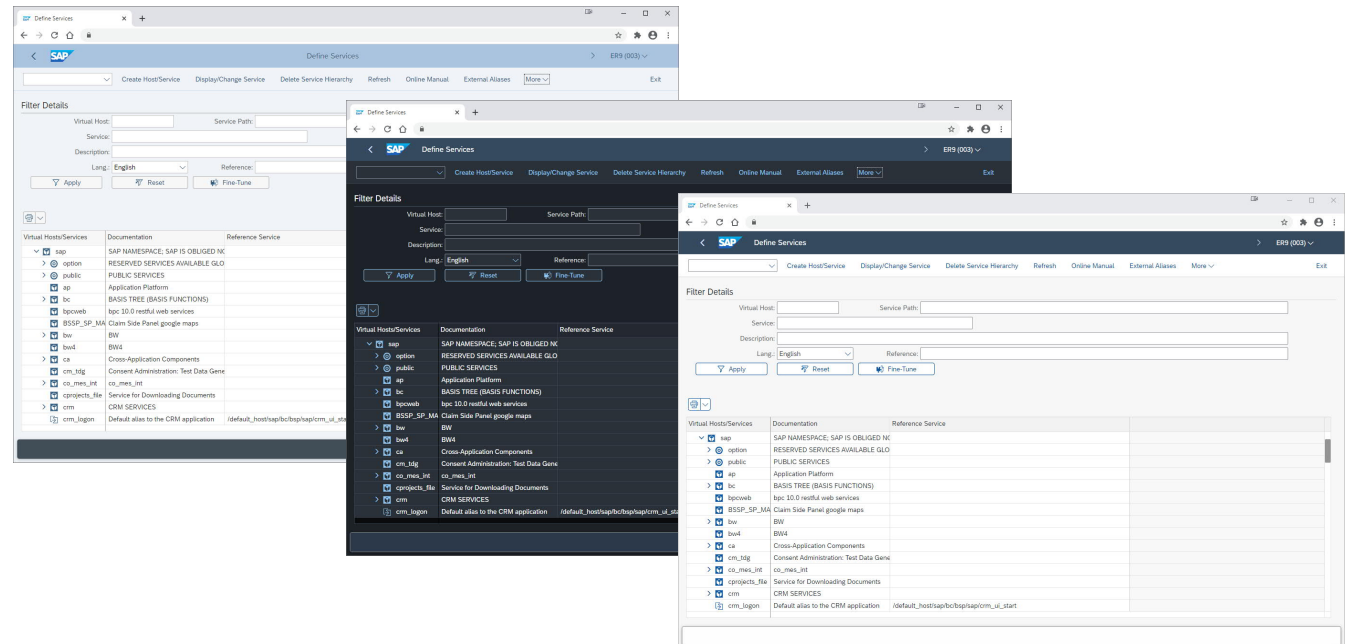
Item	Material	Order Quantity	Un	Pint	ItCa	First Date	HL Item	Overall Status	Req. Segment	Sea...	Sea...	IU...	Collec...	Theme
☐						27.10.2020						☐		
☐						27.10.2020						☐		
☐						27.10.2020						☐		

Save Cancel

What is the SAP GUI for HTML?

The SAP GUI for HTML is an user interface technology within the ABAP core system

- It extends the reach of SAP systems into the web
- It generically maps SAP Dynpro-based screens to HTML
- It has zero footprint on the client PC
- Drives transactions within the SAP system
- Can be used with all ABAP systems currently supported



What do I need to get it

Good news, everyone!

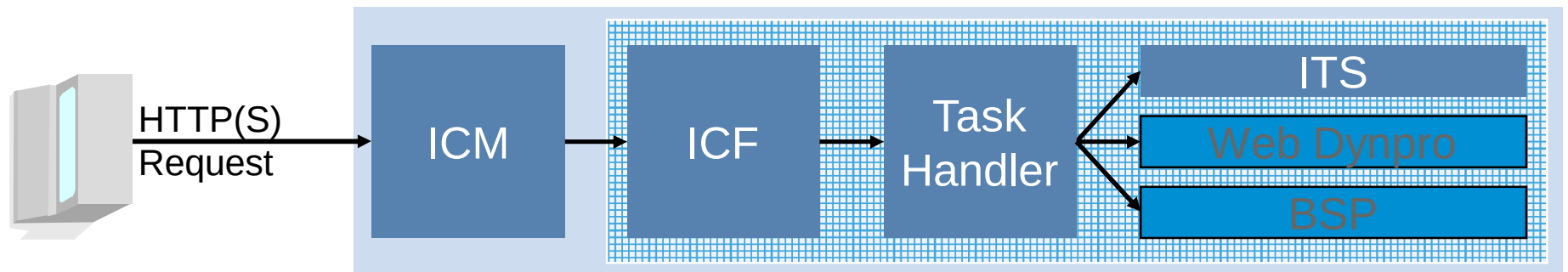
- Virtually nothing. It's already in your ABAP system
- HTTP infrastructure must be activated (ICM, ICF)
- Webgui service must be activated (TA: SICF)
- And we require a few technical services (see note 2213657 for details)

However

- If you have more than one application server you need some kind of Load-Balancing device
 - You can either use the SAP Web Dispatcher
 - Or any other HTTP Load-Balancer

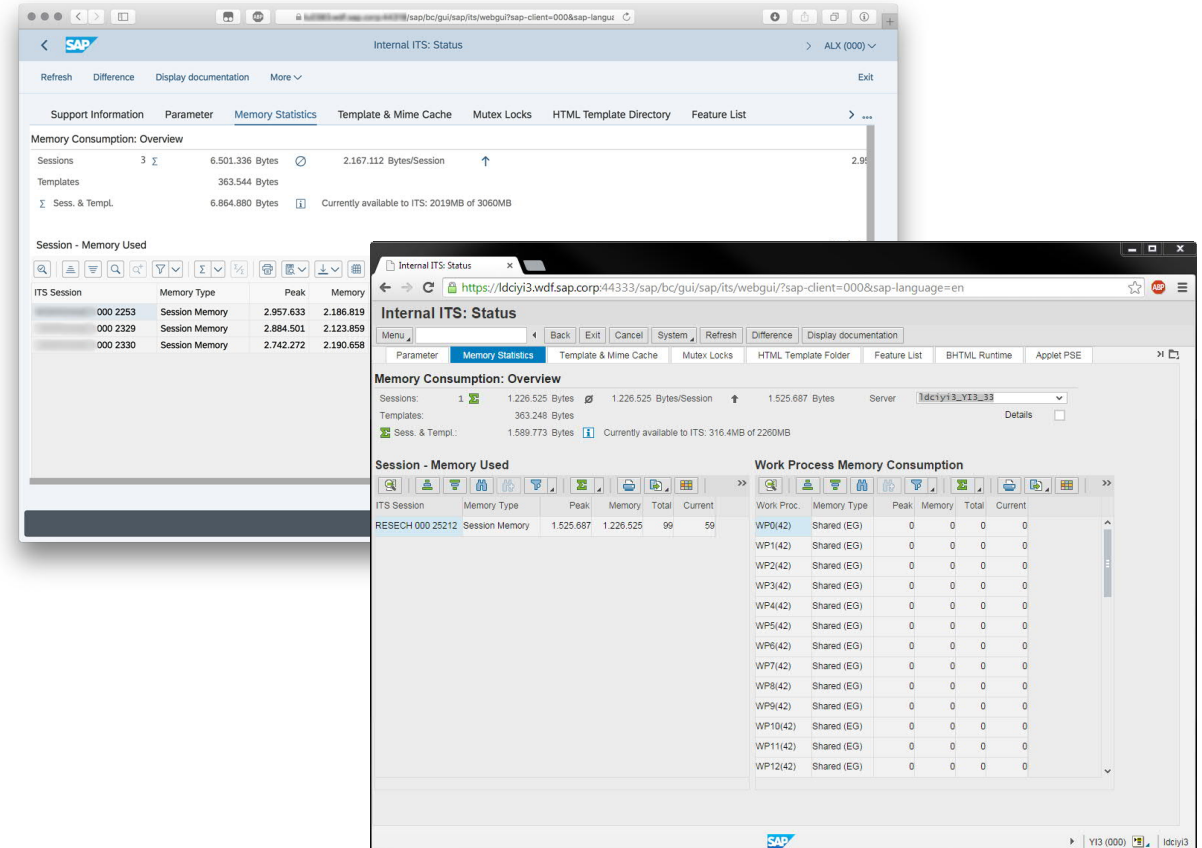
How does it work?

- You start the “Webgui” service just like any other SAP web service
- Default URL: `http(s)://host.domain/sap/bc/gui/sap/its/webgui`
- ICM is executing the responsible ICF node
- ICF calls the ITS handler
- ITS executes the called Dynpro screen and hands over the response to SAP GUI for HTML
- SAP GUI for HTML creates the HTML code needed to paint the screen in the browser and passes that code all the way back to ICF / ICM
- Browser paints the HTML page created by SAP GUI for HTML



SAP GUI for HTML: Client platforms

- **Google Chrome**
 - Current version on Windows, macOS and Android
- **Mozilla Firefox**
 - Supported versions: see note 990034
 - Windows; Linux; macOS
- **Safari**
 - macOS (by Apple supported versions only), iOS and iPadOS
- **Microsoft Edge (Chromium based version)**
 - See note 2884143
- **Internet Explorer (IE 11)**
 - By Microsoft supported versions only
 - End of support for “Quirks” mode

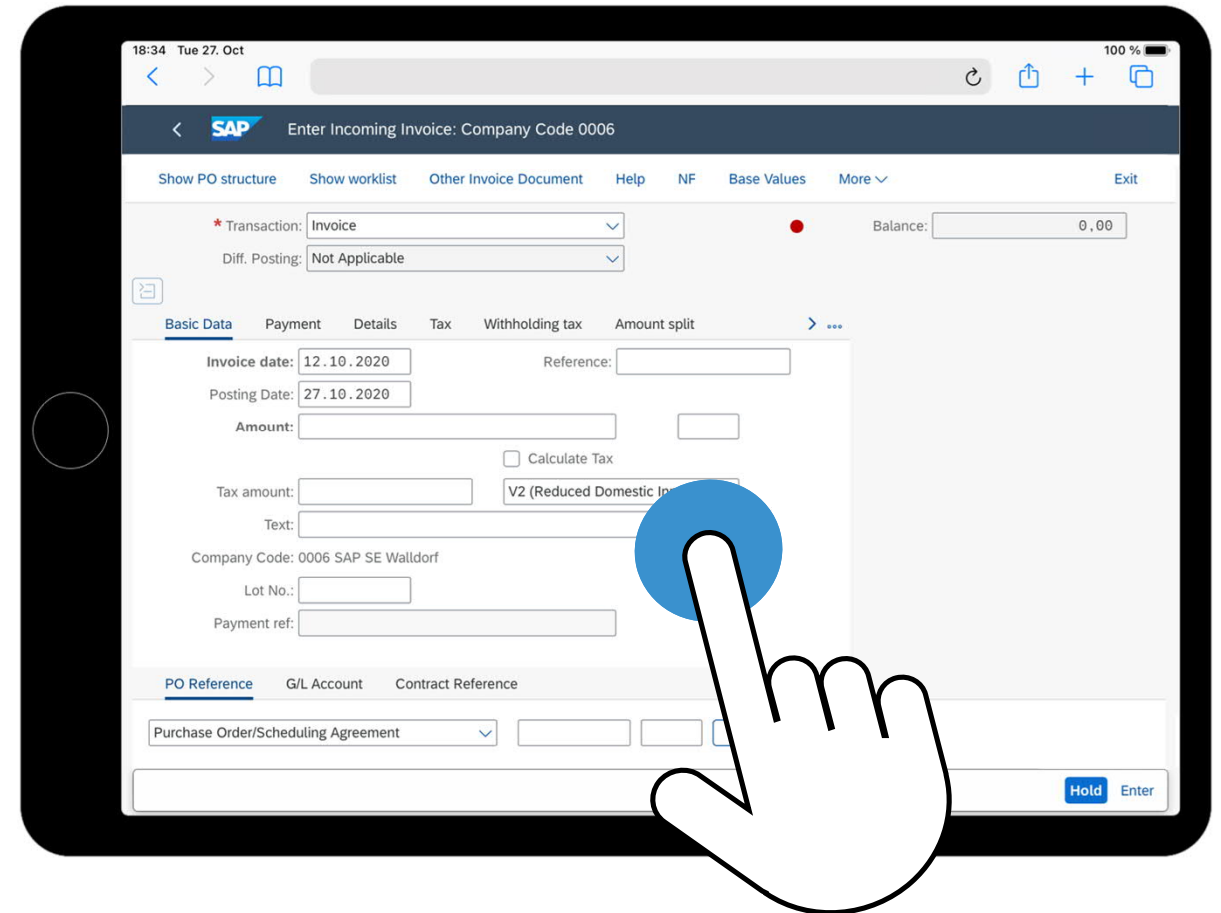


For details please check the Platform Availability Matrix: <http://support.sap.com/pam>

Running SAP GUI Transactions on Mobile Devices

SAP GUI for HTML can also be used on touch based mobile devices like Apple iPad and Android Tablets

- Development is focused on tablet computers
- Preserving existing functionality
- Mouse actions are substituted by touch gestures
- For detailed information see note 2792965



SAP GUI for HTML: Recent Innovations

Functional Improvements

- Content snag (Ctrl+Y) for ABAP list via Lasso Selection ([Note 3164701](#))
- Content snag (Ctrl+Y) for ALV grids ([Note 3164952](#))
- Unified search help aligned with Fiori design
- Support for Access Keys ([Note 3059358](#))

Visual improvements of the GUI floorplan

- 72 monospace font
- Flipped background color
- Improved colors and style in ALV grid

Control Conversion for SAP Screen Personas

- Text fields and search help fields can be converted to a dropdown list or a label

Content Security Policy

- CSP Level 3 Support

New Theme “Horizon”

Icon	Name of Icon	Comment	Length	Printable
^	ICON_NEXT_VALUE	Next value; previous entry	2	✓
v	ICON_PREVIOUS_VALUE	Previous value; next entry	2	✓
⌂	ICON_EXECUTE_OBJECT	Execute <object>	2	✓
🔍	ICON_SELECT_DETAIL	Choose <detail>; Detail	2	✓
🖨	ICON_PRINT	Print		
📄	ICON_COPY_OBJECT	Copy <object>		
⬇	ICON_SORT_DOWN	Sort in Descending Order		
⬆	ICON_SORT_UP	Sort in Ascending Order		
🔍	ICON_SEARCH	Find		
⊕	ICON_INSERT_ROW	Insert Row		
⊖	ICON_DELETE_ROW	Delete Row		
📝	ICON_CHANGE_TEXT	Text		
📝	ICON_CREATE_TEXT	Create text		
🗉	ICON_DISPLAY_TEXT	Display text	2	✓
🗉	ICON_ANNOTATION	Note; remark	2	✓
🗉	ICON_CREATE_NOTE	Create note	2	✓

	A	B	C
1	ICON_EXECUTE_OBJECT	Execute <object>	2
2	ICON_SELECT_DETAIL	Choose <detail>; Detail	2
3	ICON_PRINT	Print	2
4	ICON_COPY_OBJECT	Copy <object>	2
5	ICON_SORT_DOWN	Sort in Descending Order	2
6	ICON_SORT_UP	Sort in Ascending Order	2
7	ICON_SEARCH	Find	2
8	ICON_INSERT_ROW	Insert Row	2
9	ICON_DELETE_ROW	Delete Row	2
10	ICON_CHANGE_TEXT	Text	2

SAP Create Standard Order: Overview

Restrict Value Range (1)

Search and Select

Customers (General) v

Go Hide Filters

Search Term: Country/Region Key: Postal Code: City: Name:

Customer: Deletion Flag: Stand. address: Valid On:

Items (500)

SearchTerm	C/R	PostalCode	City	Name 1	Customer	DelF	Partner
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000010		
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000012		CCSC000012
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000016		CCSC000016
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000017		CCSC000017
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000018		CCSC000018
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000019		CCSC000019
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000020		CCSC000020
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000021		CCSC000021
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000022		CCSC000022
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000023		
CCS	DE	69190	WALLDORF	TEST CUSTOMER: CCS	CCSC000024		

More than 500 input options

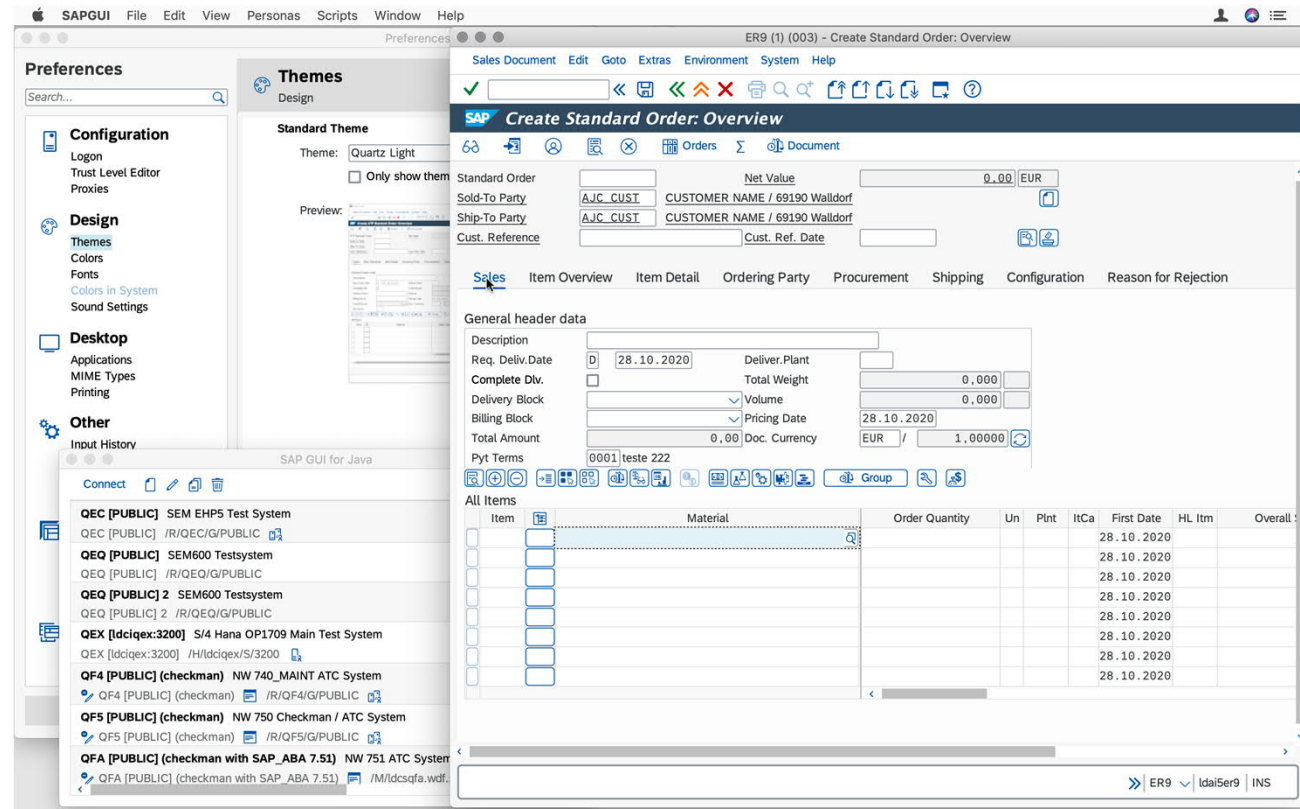
OK Cancel

Limitations of the SAP GUI for HTML

Because of the technical attributes of HTML or the Web Browser, the use of SAP GUI for HTML is somewhat restricted as compared to the use of a GUI, which works on operating system level.

- The controlling of an OLE automation server is not possible
- Keyboard navigation is limited as known from other web applications
- Office integration is view only
- Some controls (like the advanced ABAP Editor Control) in SAP GUI for Windows used for special purposes are not available in SAP GUI for HTML
- To see if a transaction is supported for use with SAP GUI for HTML check in transaction SE93
- Anyway most of the transactions work perfectly
- For detailed information see note 314568

SAP GUI for Java



What is the SAP GUI for Java?

The SAP GUI for Java is unified SAP front end running on Linux, macOS and Windows

- The technology behind is a combination of Java and C/C++ running with the same codebase on all platforms
- All rendering is done in Java using Swing and JavaFX components to provide highly portable and consistent functionality on all supported platforms
- The corresponding components of the standard ActiveX controls used in SAP GUI for Windows are reimplemented as JavaBeans
- Already existing C/C++ libraries on the respective platforms are used for functions such as network communication, protocol handling or RFC
- It can be used with all SAP releases currently supported

What's new in SAP GUI for Java 7.80

Enhancements and New Features

- Support for New OS Versions:
 - Fedora 36
 - openSUSE 15.4
 - Red Hat Enterprise Linux 9
 - macOS Ventura
- Support for Java SE 17:
SapMachine 17 with OpenJFX 17
- New About Dialog and App Icons
- New sound schemes for audio feedback.

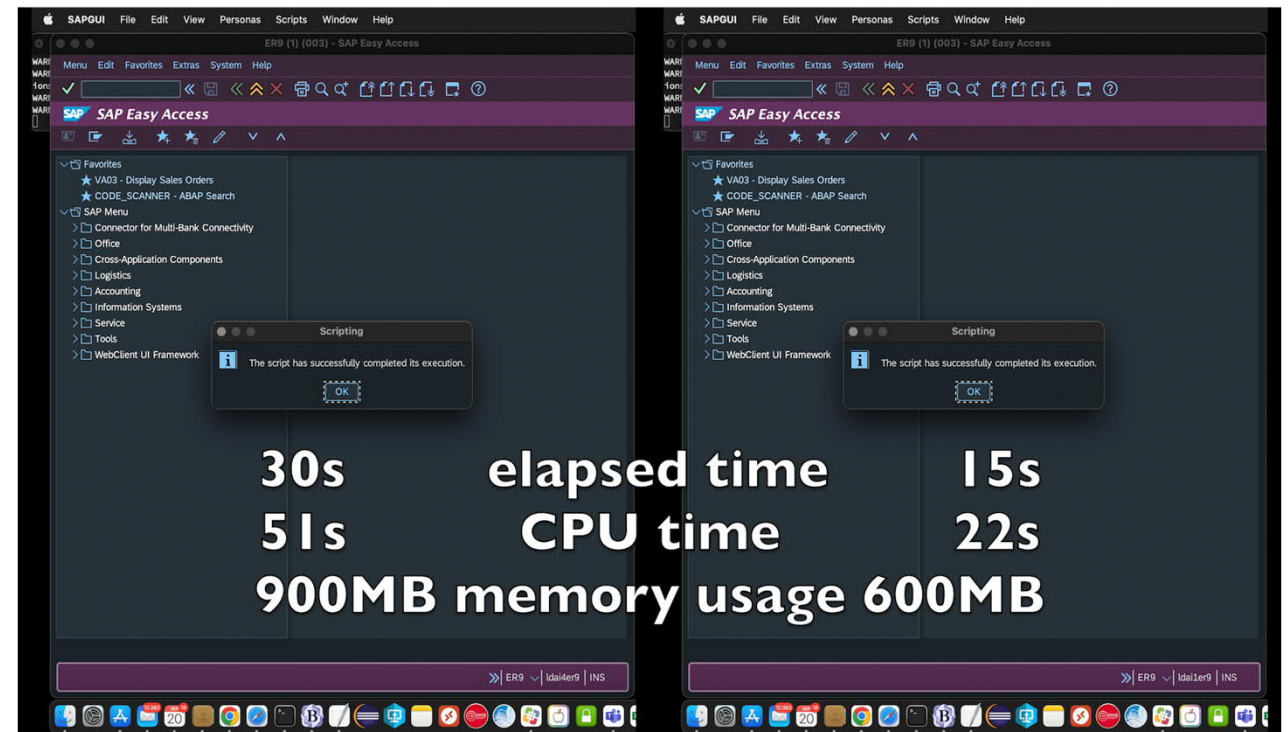
Discontinued Features

- Belize Theme
- Connections to Web Dynpro Backends

Details see SAP note [3204106](#)

- Support for New Platforms:
Native support for macOS on Apple silicon

Comparison Intel (left) and native ARM (right) version on M1 Max with startup, connect and displaying various application screens



Supported operating systems

- SUSE Linux Enterprise Desktop 15
- SUSE Linux Enterprise Server 15
- openSUSE Leap 15.3
- Fedora 36
- Red Hat Enterprise Linux 8 and 9
- Ubuntu 22.04
- macOS Monterey
- macOS Ventura
- Windows 10 64-bit (Professional and Enterprise Edition)
- Windows 11 Enterprise

Newest SapMachine 17 and OpenJFX 17 included in installer

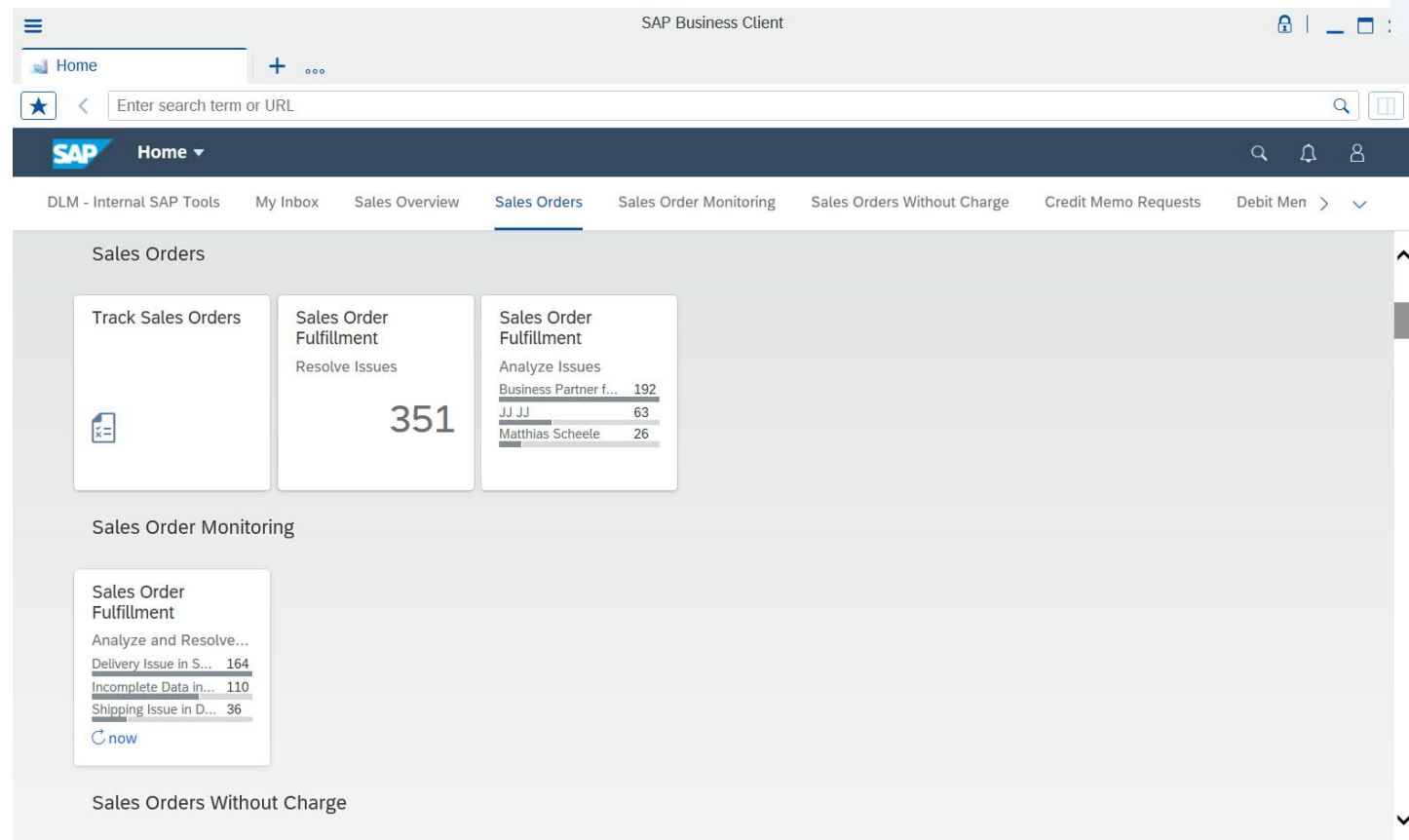
Detailed requirements regarding supported operating systems for SAP GUI for Java are documented in SAP note [3204095](#).

Limitations of the SAP GUI for Java

Limitations compared to SAP GUI for Windows

- Office integration into SAP transactions is not available
- Some GUI components like the BEx explorer are not available
- Some controls in SAP GUI for Windows used for special purposes are not available as JavaBean in SAP GUI for Java, for example
 - SAP SmartForms
 - advanced ABAP Editor Control
- Graphical screen painter is not available
- To see if a transaction is supported for use with SAP GUI for Java check in transaction SE93
- Anyway most of the transactions work perfectly

SAP Business Client



The screenshot displays the SAP Business Client web application. The top navigation bar includes the SAP logo, a 'Home' dropdown, and search, notification, and user icons. Below this, a secondary navigation bar lists various modules: 'DLM - Internal SAP Tools', 'My Inbox', 'Sales Overview', 'Sales Orders' (which is the active module), 'Sales Order Monitoring', 'Sales Orders Without Charge', 'Credit Memo Requests', and 'Debit Men'. The main content area is titled 'Sales Orders' and contains three primary cards. The first card, 'Track Sales Orders', features a document icon. The second card, 'Sales Order Fulfillment', shows a large '351' and the text 'Resolve Issues'. The third card, also titled 'Sales Order Fulfillment', includes the text 'Analyze Issues' and a table of data. Below these cards is a 'Sales Order Monitoring' section with a card titled 'Sales Order Fulfillment' that lists specific issues and their counts. At the bottom, a section for 'Sales Orders Without Charge' is partially visible. A small blue hummingbird icon is positioned in the top right corner of the image.

Sales Order Fulfillment	
Business Partner f...	192
JJ JJ	63
Matthias Scheele	26

Sales Order Fulfillment	
Delivery Issue in S...	164
Incomplete Data in...	110
Shipping Issue in D...	36

SAP Business Client 8.00



- **Inplace Navigation:** Using a FLP connection, it's now possible to navigate from HTML based content to native SAPGUI for Windows and vice versa within the same tab.
- **Separate Theming:** The Shell Window and System Selection Screen can be themed separately.
- **History:** With 8.00, the Business Client offers a full-blown history across web-based and native SAPGUI for Windows navigations.
- **Sap WorkZone Integration:** Similar to the Launchpad Service, SAP WorkZone can be used within the Business Client. Wherever possible, SAPGUI transactions are rendered natively via SAPGUI for Windows.
- **Favorites Export & Import:** Favorites can be exported from one user and are then available to other users as well.
- **Cef Chromium Removal:** With the Microsoft Edge browser control already integrated, the Business Client offers a mature Chromium-based browser control with better supportability and easier maintenance.

SAP GUI for Windows and SAP Business Client in S/4HANA on premise



Home EPM Sales Order 050... NCP (1) 120

Enter search term or URL

SAP EPM Sales Order 0500000010

Change More

Exit

Header Data Items

Business Partner

Business Partner ID: 0100000002

Company DelBont Industries

Street 1 2345 King Street 1

Postal Code/City 19899 Wilmington, Delaware

Country US USA

Lifecycle

Status

☐ Confirmed

☒ New

☐ Closed

☐ Cancelled

Delivered

Status

☐ Yes

☒ No

Paid

Status

☐

Amounts

Net Amount	695,75	EUR
Tax Amount	132,20	EUR

Note

EPM DG: SO ID 0500000010 Deliver as fast as possible

Sales Orders

Track Sales Orders

Sales O Fulfillment

Resolve I

Sales Order Monitoring

Sales Order Fulfillment

Analyze and Resolve...

Delivery Issue in S... 164

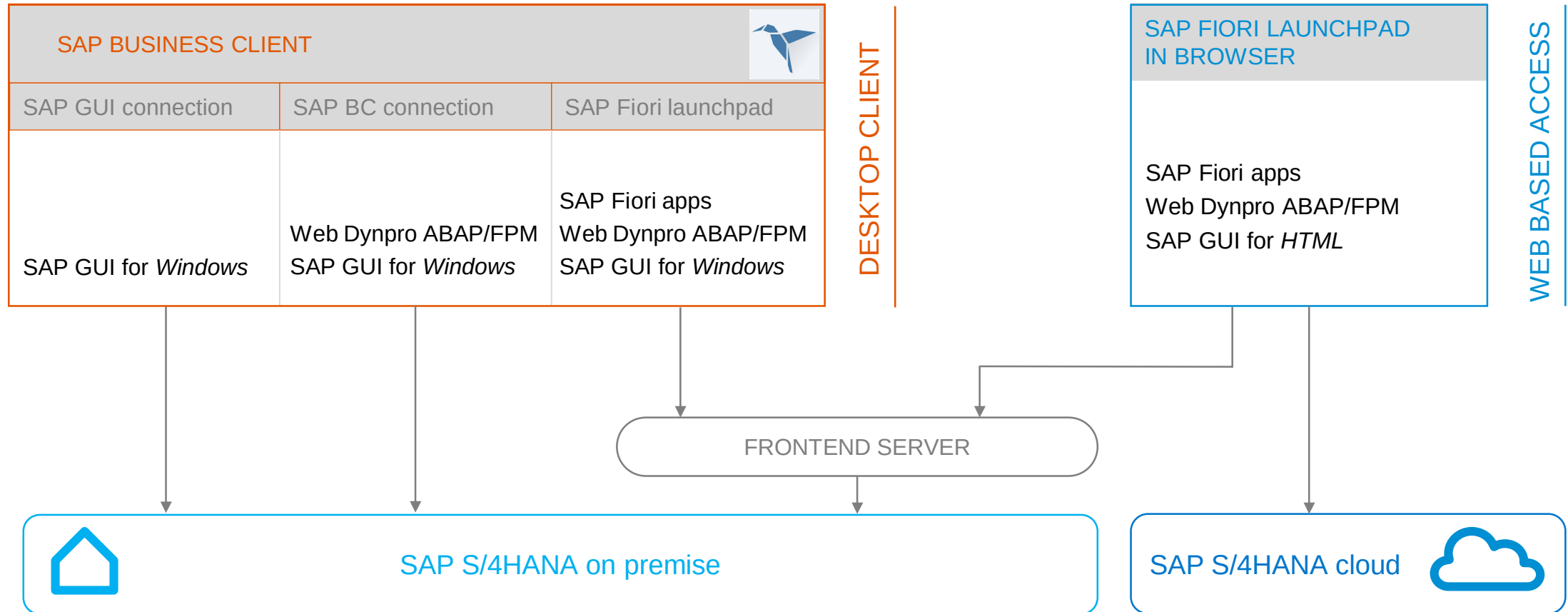
Incomplete Data in... 110

Shipping Issue in D... 36

now

Sales Orders Without Charge

Client Options for SAP S/4HANA

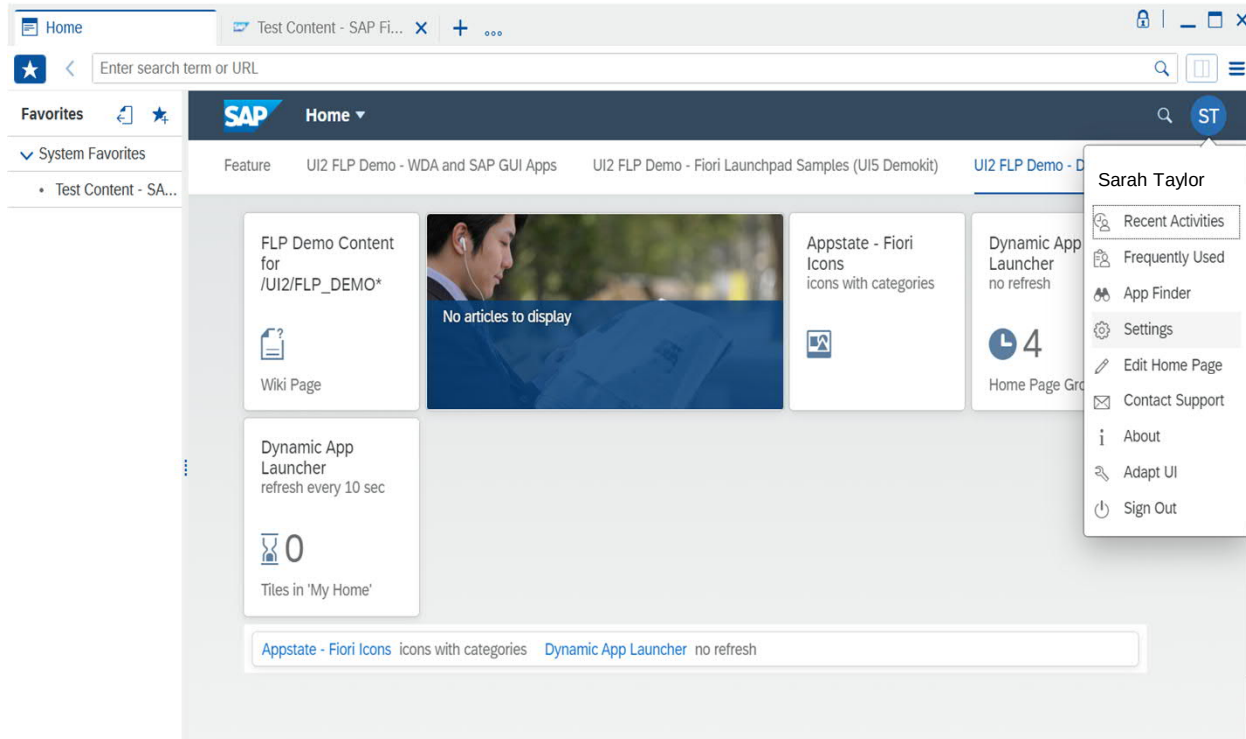


This is the current state of planning and may be changed by SAP at any time.

SAP Business Client – Embedding Scenarios

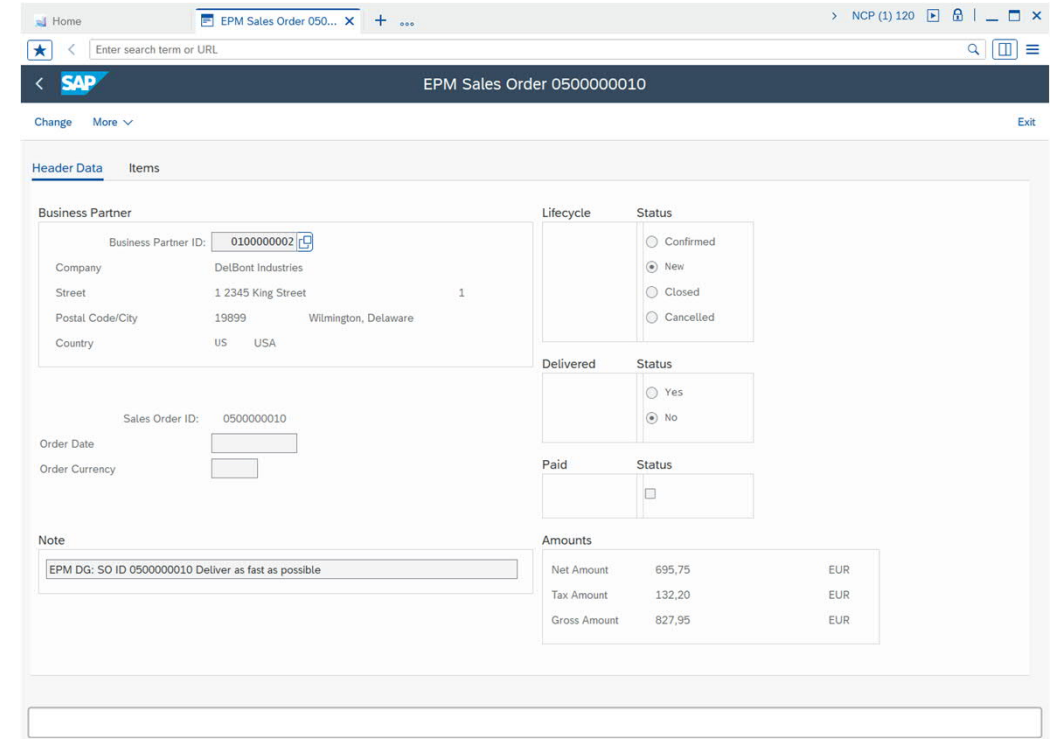
Fiori launchpad and Fiori apps on dedicated tab

- Full shell header including SAP Fiori Launchpad functionality (e.g. “Me” area, branding, search, etc.)



SAP GUI for Windows

- Minimal shell header including “Back” button, branding and title
- For other functions switch to FLP tab



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